

Genome-wide identification of BURP domain-containing genes in rice reveals a gene family with diverse structures and responses to abiotic stresses

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Abstract Increasing evidence suggests that a gene family encoding proteins containing BURP domains have diverse functions in plants, but systematic characterization of this gene family have not been reported. In this study, 17 *BURP* family genes (*OsBURP01–17*) were identified and analyzed in rice (*Oryza sativa* L.). These genes have diverse exon–intron structures and distinct organization of putative motifs. Based on the phylogenetic analysis of BURP protein sequences from rice and other plant species, the BURP family was classified into seven subfamilies, including two subfamilies (BURP V and BURP VI) with members from rice only and one subfamily (BURP VII) with members from monocotyledons only. Two *BURP* gene clusters, belonging to BURP V and BURP VI, were located in the duplicated region on chromosome 5 and 6 of rice, respectively. Transcript level analysis of *BURP* genes of rice in various tissues and organs revealed different tempo-spatial expression patterns, suggesting that these genes may function at different stages of plant growth and development. Interestingly, all the genes of the BURP VII subfamily

were predominantly expressed in flower organs. We also investigated the expression patterns of *BURP* genes of rice under different stress conditions. The results suggested that, except for two genes (*OsBURP01* and *OsBURP13*), all other members were induced by at least one of the stresses including drought, salt, cold, and abscisic acid treatment. Two genes (*OsBURP05* and *OsBURP16*) were responsive to all the stress treatments and most of the *OsBURP* genes were responsive to salt stress. Promoter sequence analysis revealed an over-abundance of stress-related *cis*-elements in the stress-responsive genes. The data presented here provide important clues for elucidating the functions of genes of this family.

Keywords Abiotic stress · BURP domain · *Oryza* · RD22 · Transcript level

Abbreviations

ABA Absciscic acid
ABRE ABA responsive element
BURP BNM2, USP, RD22, and PG1 β
DRE Dehydration-responsive element
LTRE Low temperature-responsive element

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Introduction

The BURP domain is a conserved amino acid motif whose name is based on four members: BNM2, USP, RD22, and PG1 β (Hattori et al. 1998). BURP domain-containing proteins consist of several conserved modules: an N-terminal hydrophobic domain with a putative signal peptide; a BURP domain containing two conserved cysteine (C) residues and four repeat cysteine–histidine (CH) motifs in the C-terminal; and a variable internal region containing a short

conserved segment or a short segment plus an optional segment consisting of repeated units, which is unique to each member (Hattori et al. 1998). Many BURP domain-containing proteins have been reported to be ubiquitous in plant species. Examples are BNM2, an oilseed rape (*Brassica napus* L.) microspore protein (Boutilier et al. 1994; Treacy et al. 1997); VfUSP, an unknown seed protein from faba bean (*Vicia faba* L.) (Bassüner et al. 1988; Baumlein et al. 1991); RD22, an *Arabidopsis thaliana* (L.) Heynh. dehydration-responsive protein (Yamaguchi-Shinozaki and Shinozaki 1993); PG1 β , a β -subunit of polygalacturonase isoenzyme 1 of tomato (*Lycopersicon esculentum* L.) (Zheng et al. 1992; Watson et al. 1994); ADR6, a soybean [*Glycine max* (L.) Merr.] auxin down-regulated protein (Datta et al. 1993); SALI3-2, an aluminum-induced soybean protein (Ragland and Soliman 1997); ASG1, an apomixis-specific gene from guinea grass (*Panicum maximum* Jacq.) (Chen et al. 2005); and SCB1, seed coat BURP-containing protein 1 from soybean (Batchelor et al. 2002). The BURP domain has been found in abundant plant proteins as mentioned earlier. Even in the ancient moss *Physcomitrella patens* (Hedw.) Bruch and Schimp. in B.S.G., BURP domain-containing proteins were identified (Rensing et al. 2008). So far, no BURP domain-containing protein has been identified in a nonplant species.

Based on phylogenetic analysis of the putative BURP domain-containing proteins in soybean and other plant species, the members of the BURP family can be classified into BNM2-like, USP-like, RD22-like, and PG1 β -like subfamilies (Granger et al. 2002). The major variability among the members of the BURP family occurs in the region between the hydrophobic signal sequence and the BURP domain. This region includes a short conserved segment and an optional segment consisting of repeated units. The repeated units are absent in the BNM2-like proteins. The PG1 β -like proteins can be distinguished from the members of other subfamilies by the presence of a sequence segment containing a 14 amino acid sequence, for instance, FTNYGXXGNGGXXX (where X = any amino acid residue) in PG1 β (Zheng et al. 1992). Both the USP-like and RD22-like proteins can be distinguished from other subfamilies by a sequence segment containing ~30 amino acids, but the RD22-like proteins have a segment consisting of repeated units of approximately 20 amino acids while the USP-like proteins do not contain such a segment (Granger et al. 2002).

Even though the BURP family can be easily classified based on the sequence features, the functions of most members in this family are largely unknown. Previous reports suggested that the BURP family may play important roles in different aspects of plant vegetative and reproductive development. PG1 β , the second noncatalytic subunit of the PG1 complex identified from tomato fruit, is involved in

the immobilization of PG2 and regulation of its activity (Watson et al. 1994). The PG1 complex plays a significant role in regulating pectin metabolism by limiting the extent of pectin solubilization and depolymerization during fruit ripening in tomato (Zheng et al. 1992; Watson et al. 1994). SCB1 (seed coat BURP-domain protein 1), the mRNA of which is detected specifically within the seed coat during the early stages of the soybean seed development, was suggested to be involved in the formation of the soybean seed coat by governing the differentiation of the seed coat parenchyma cells (Batchelor et al. 2002). OsRAFTIN1, an anther-specific BURP protein in rice (*Oryza sativa* L.), is exclusively expressed in the tapetum during postmeiotic stages when a young microspore undergoes rapid expansion, and is essential for male fertility. When the *OsRAFTIN1* gene is suppressed, the mature anthers become nondehiscent and shortened and the pollen grains collapse. OsRAFTIN1 has been detected in the tapetum, Ubisch bodies, locule, and microspore, so it may play an important role in transport of sporopollenin from the tapetum to developing microspores via Ubisch bodies (Wang et al. 2003). RA8, another anther-specific BURP protein, is expressed specifically in the tapetum, endothecium, and connective tissue, but not in pollen grains, starting from the tetrad stage and reaching the maximum level at the late vacuolated-pollen stage. RA8 is thus suggested to play an important role in microspore development and dehiscence of anther (Jeon et al. 1999).

Quite a few members of the BURP family have been reported as responsive to stress treatments. RD22, a drought-responsive gene encoding a BURP protein, is induced by drought mediated by abscisic acid (ABA) signaling. The responsiveness of RD22 to drought requires protein biosynthesis for ABA-dependent gene expression (Yamaguchi-Shinozaki and Shinozaki 1993). The expression of RD22 was found to be controlled by the cooperation of rd22BP1 and AtMYB2 proteins (Abe et al. 1997, 2003). The transcript level of *BnBDC1*, a shoot-specific gene encoding a BURP protein in oilseed rape, was up-regulated by salt, ABA, and osmotic stresses and down-regulated by salicylic acid; thus it was suggested to be involved in both abiotic and biotic stresses (Yu et al. 2004). Four genes from *Bruguiera gymnorhiza* (*BgBDC1*, 2, 3, and 4) encoding BURP proteins with sequence similarity to RD22, were induced by salt, ABA, and drought stresses (Banzai et al. 2002). The expression of *SALI3-2* and *SALI5-4a*, 2 putative BURP protein genes, were induced by aluminum stress in soybean (Ragland and Soliman 1997). The over-expression of *SALI3-2* can elevate the salt tolerance of yeast, but signal peptide deletion weakened the salt tolerance (Tang et al. 2007). These reports suggest that BURP family genes may be crucial not only for plant development but also for response and adaptation to stresses.

Rice is one of the most important crops and a model plant for monocot species. However, only two BURP protein-encoding genes, *OsRAFTINI*, and *RA8*, have been reported (Jeon et al. 1999; Wang et al. 2003). Even though the rice genome has been sequenced for a few years, a genome-wide analysis of the BURP family in this species has not been reported. In this study, 17 putative genes of the BURP family were identified in the rice genome and these genes were classified into five subfamilies. Toward a long-term goal of revealing the functions of all members in this family, transcript expression levels of this family were investigated in various rice tissues and organs from different developmental stages and in the seedling leaves under various abiotic stresses. The results presented in this study provide an important reference for function studies of BURP family genes in rice.

Materials and methods

Identification and sequence analysis of *BURP* family genes in rice

The Hidden Markov Model profile of BURP domain (PF03181), which was generated by alignments of 14 seed sequences, was downloaded from the Pfam (Finn et al. 2006) (<http://pfam.sanger.ac.uk/Software/Pfam>). This domain was used to identify all BURP domain-containing proteins (designated *OsBURP*) in rice genome by searching the domain sequence against the rice annotation databases including the *japonica* genomic sequence and annotation database in TIGR (The Institute for Genomic Research) (<http://rice.plantbiology.msu.edu>; Ouyang et al. 2007) and the *japonica* full length cDNA date of KOME (Knowledge-based *Oryza* Molecular biological Encyclopedia) (<http://red.dna.affrc.go.jp/cDNA>). The conserved sequences of BURP proteins were used to search the database at NCBI and TAIR (The Arabidopsis Information Resource) (<http://www.arabidopsis.org/>; Garcia-Hernandez et al. 2002) by BLAST to retrieve the putative BURP homologs in other species. The Pfam (Finn et al. 2006) (<http://pfam.sanger.ac.uk>) and SMART (<http://smart.embl-heidelberg.de>; Letunic et al. 2004) databases were used to confirm each predicted BURP protein sequence as a member of the BURP family (i.e., containing a BURP domain). The position of each *OsBURP* gene in the rice chromosome was determined by BLAST search against the rice genomic sequence of each chromosome at the Gramene database (<http://www.gramene.org>; Ware 2007). Through comparison of the full-length cDNA or predicted coding sequence (CDS) with their corresponding genomic sequence, the exon–intron organizations of the *OsBURP* genes were determined.

To identify the conserved motifs in BURP proteins, online MEME (Multiple EM for Motif Elicitation) was employed to analyze the protein sequences of BURP members from different plant species with an expected value lower than 2×10^{-30} (<http://meme.sdsc.edu>; Bailey et al. 2006). The obtained motifs were annotated by searching the motif sequences against InterPro (<http://www.ebi.ac.uk/interpro/>) and NCBI-CD databases (<http://www.ncbi.nlm.nih.gov/Structure/cdd/wrpsb.cgi>). The SMART search program was used to confirm the BURP domain and to obtain the protein sequences of the BURP domain, and the protein sequences of BURP domain were in turn used to generate an alignment by ClustalX 1.83 for phylogenetic analysis (Thompson et al. 1997; Letunic et al. 2004). The phylogenetic tree was constructed using PHYLIP 3.63 (Retief 2000), through a series of programs including seqboot, protdist, maximum-likelihood, and consense. The MEGA 3.0 software (Kumar et al. 2004) was used to obtain a readable form of the phylogenetic tree.

To analyze putative *cis*-acting regulatory DNA elements (*cis*-elements) in the promoters of the rice BURP genes, online PLACE 26.0, a database of nucleotide sequence motifs found in plant *cis*-acting regulatory DNA elements, was employed to analyze 1,000 bp sequences upstream of the 5' end of the full-length cDNAs or predicted CDS extracted from genomic sequences (<http://www.dna.affrc.go.jp/PLACE/index.html>; Higo et al. 1999).

Plant materials and treatments

Seeds of Minghui 63 (*O. sativa* L. ssp. *indica*, a commercialized parent line for hybrid rice in China) were germinated in plates with water. Plants were grown in a greenhouse with a 14 h light/10 h dark cycle at 24–28°C and various tissue and organ samples were collected at seedling and reproductive stages.

For transcript level analysis of *OsBURP* genes under stress, Minghui 63 plants were planted in plastic trays filled with sandy soil and placed in the greenhouse. Three-week-old seedlings were prepared for abiotic stress treatments mainly according to Xiong and Yang (2003). Drought stress was induced by withholding water from the trays, and seedling leaves were sampled at normal (before withholding water), slight drought [with the average relative water content (RWC) at about 90%], moderate drought (about half of leaves rolled, RWC at about 83%), and severe drought (all leaves completely rolled, RWC at about 70%) conditions. Seedling roots were submerged in 200 mM NaCl solution for salt stress, and seedling leaves were sampled at 0, 1, 6, and 12 h after the treatment. For ABA treatment, seedling leaves were sprayed with 0.1 mM ABA solution and sampled at 0, 10, 30 min, and 3 h after treatment. For cold stress, the seedlings were transferred

into a growth chamber at 4°C and sampled at 0, 1, 3, and 10 h after treatment.

RNA extraction and real-time polymerase chain reaction (PCR) analysis

Total RNAs of the collected samples were extracted using TRIzol reagent (Invitrogen). For real-time PCR analysis, first-strand cDNA was synthesized using Superscript II reverse transcriptase (Invitrogen) according to the manufacturer's instruction. Real-time PCR was conducted on an ABI PRISM 7500 real-time PCR system (Applied Biosystems). The gene-specific primers for real-time PCR analysis were designed by Primer Express Version 2.0 (Applied Biosystems). The expression level of rice *Actin1* gene (accession no. X16280) was used as the internal control with primers 5'-TGGCATCTCTCAGCACATTCC-3' and 5'-TGCACAATGGATGGGTCAGA-3'. Real-time PCR amplifications were performed in an optical 96-well plate. Each reaction was done in a volume of 25 µL containing 12.5 µL of 2 × SYBR green master reagent (Applied Biosystems), 1.0 µL diluted transcription product, and 0.2 µL each gene-specific primer. The thermal cycle was used as follows: 95°C for 3 min, 40 or 45 cycles of 95°C for 30 s, 60°C for 30 s, and 72°C for 1 min. Dissociation curve analysis was performed as follows: 95°C for 15 s, 60°C for 20 s, and 95°C for 15 min. The relative expression levels were determined as described previously (Liang et al. 2006). The efficacies of amplification of *OsBURP* gene primers were analyzed by using the LinRegPCR program with liner regression analysis (Ramakers et al. 2003). If an efficacy of amplification was less than 2, the result was proofread.

Results

Identification of *BURP* genes in rice

To identify all putative genes of the BURP family in the rice genome, we used BLAST to search the rice genome annotation database (TIGR, release 5.0) and the full-length cDNA database (KOME) with the Hidden Markov model Profile of BURP domain as a query. Through this approach, 17 putative *BURP* genes, designated *OsBURP01* to *OsBURP17*, were identified, including the reported *OsRAFTIN1* and *RA8* genes (corresponding to *OsBURP13* and *OsBURP15*, respectively). The information of BURP family genes in rice, including accession numbers of the full-length cDNAs and the corresponding BAC/PAC clones and their physical locations on chromosomes, are listed in Table 1. The 17 *OsBURP* genes are distributed among all the chromosomes except chromosomes 3, 7, and 12. The

distribution of the genes was uneven: five genes on chromosome 5, three genes on chromosome 6, two genes on both chromosome 8 and 9, and one gene each on the other five chromosomes. Interestingly, the five genes (*OsBURP01*, *OsBURP02*, *OsBURP06*, *OsBURP07*, and *OsBURP08*) on chromosome 5 constitute a gene cluster with high sequence similarity to each other and 4 of them (*OsBURP01*, *OsBURP02*, *OsBURP06*, and *OsBURP07*) are very closely located to each other in the same BAC (AC108498) (Fig. 1a). Another gene cluster consists of three genes (*OsBURP09*, *OsBURP10*, and *OsBURP11*) on chromosome 6. Currently, nine putative *OsBURP* genes are supported by full-length cDNAs in the KOME database. However, one base pair (bp) is absent in the cDNA sequence of AK120942 compared to the predicted cDNA sequence from TIGR database for *OsBURP13*, and 43 amino acids are missing in the predicted protein sequence of the cDNA AK067523 compared to the sequence from TIGR database for *OsBURP17*. Diverse exon–intron structures were identified by comparing the full-length cDNAs or predicted CDS with the genomic sequence of *OsBURP* genes in the TIGR or GenBank database (Fig. 1b). The result showed that all *OsBURP* genes except *OsBURP06* and *OsBURP12* contain introns.

Sequence analysis of BURP proteins

All the *OsBURP* protein sequences were subjected to search against SMART database and the result showed that the signal peptide was present in 12 *OsBURP* proteins but absent in the other five members (*OsBURP04*, *OsBURP05*, *OsBURP06*, *OsBURP08*, and *OsBURP11*). Except *OsBURP01* and *OsBURP15*, all other members of the family have a complete BURP domain.

To study the evolutionary relationships of BURP proteins in plants, a phylogenetic tree was generated using the protein sequences of 38 putative BURP-domain sequences, including all the 17 members from rice and 21 members selected from other plants (*Arabidopsis*, mangrove, tomato, soybean, bean [*Vicia faba*], oilseed rape, and cotton [*Gossypium hirsutum* L.]). Based on the domain composition and phylogenetic relationship of the protein sequences, the 38 plant BURP proteins can be divided into 7 subfamilies: BNM2-like, USP-like, RD22-like, PG1β-like, BURP V, BURP VI and BURP VII (Fig. 2). Among the seven subfamilies, four subfamilies (BNM2-like, USP-like, RD22-like, and PG1β-like) have been reported. The other three subfamilies, designated as BURP V, BURP VI, and BURP VII, respectively, were considered as new subfamilies since the optional repeated sequences were absent in all these three subfamilies and the distances between the second and the third CH were longer in the BURP VI subfamily than in other subfamilies. Except the USP-like and BNM2-like

Table 1 Basic information and primer sequences of real-time PCR analysis of BURP family genes in rice

Gene ^a	Accession no.		Genomic locus ^d		Exon ^e	Chr ^f	Primers for real-time PCR		PCR eff ^g
	TIGR ^b	KOME ^c	Accession no.	Mb			Forward primers	Reverse primers	
<i>OsBURP01</i>	LOC_Os05g12400	AK060149	AC108498	7.11	2	5	GGTGGCAAAGCTGAATTAATGAA	GGAAGGAGCATGCTAAATAAATTAGC	2.002
<i>OsBURP02</i>	LOC_Os05g12640	AK104640	AC108498	7.23	2	5	CATTGCGGACCTCATCAGC	CCGTATCCTCTTATCCTTCGA	2.012
<i>OsBURP03</i>	LOC_Os01g53240	AK106043	AP003560	30.9	3	5	TTTGGTCTTTGTATTGGATGGT	ACCCGCATACTAAAACTATACATCAAAC	2.058
<i>OsBURP04</i>	LOC_Os02g18690	NA	AP004168	10.9	6	2	CAGCGCGGGATGCTCTT	AGCTTCGTGCTTCCGG	2.008
<i>OsBURP05</i>	LOC_Os04g14990	NA	AL731592	8.4	3	4	GCGTCGCAGCAGTTCTTCA	AATACTTTTGCTGTTTGAGGTT	2.011
<i>OsBURP06</i>	LOC_Os05g12410	NA	AC108498	7.12	1	5	GAGCTATGTGCGTATCATGTCTTATCA	TGCCACACGTATAGTGGCATGT	1.754
<i>OsBURP07[*]</i>	OJ1076_H08.14	NA	AC108498	7.23	2	5	TGCCACAGGACGACATGCT	CAATTTTATCCCTTTACGC	2.018
<i>OsBURP08</i>	LOC_Os05g13490	NA	AC135422	7.46	2	5	TCTTTATATCGGTTAGCTAGCACAGAA	TCTTCCAGTACAGCTCCGCC	2.063
<i>OsBURP09</i>	LOC_Os06g13240	NA	AP004994	7.28	5	6	GTGTCCTCGCTAGGCACAAAG	GCACACCTTCCATTGGAGCT	2.037
<i>OsBURP10</i>	LOC_Os06g17000	AK064437	AP005695	9.86	3	6	CGGTATACAAATCGGCGTTGG	CGCACGTTCTTCTCCCTTG	2.041
<i>OsBURP11[*]</i>	OSINBa0042E12.9	AK106070	AP005748	11.3	8	6	GGAATGGATCATGGACGCC	GATGACGGAGCCCGGC	2.025
<i>OsBURP12</i>	LOC_Os08g29200	AK101838	AP004675	17.8	1	8	GGCTCCACGGCTGG	ACCATCAACACCGGTAACATTG	2.023
<i>OsBURP13</i>	LOC_Os08g38810	AK120942	AP000364	24.4	3	8	TGGGCCAAGAACGTGAAGAG	TAGCTCACAAATGCCGTCGC	2.030
<i>OsBURP14</i>	LOC_Os09g16010	NA	AP005863	9.78	2	9	ACCGTCCGCGTGTACGAG	CCACGCCATGGTTGATCTTTC	2.003
<i>OsBURP15</i>	LOC_Os09g30320	NA	AP005392	18.5	3	9	CGATGAACCTTCAACTACGACGACT	ACGCGGTTCAATGCATCC	2.031
<i>OsBURP16</i>	LOC_Os10g26940	AK120091	AC027661	13.8	2	10	TTGTGGTTCGGAGTACCGTACTG	TCCCTTGACCTTACCAATCATGA	2.037
<i>OsBURP17</i>	LOC_Os11g06980	AK067523	AC128643	3.46	3	11	CTACGAACTCCAAGTTCTCTGCC	CCAGGATCTCGGTAGTCTTTAG	2.033
<i>Actin1</i>	LOC_Os03g50885	AK100267					TGGCATCTCTCAGCACATTCC	TGCACAAATGGATGGGTCTAGA	2.000

^a Systematic nomenclature of rice *BURP* genes

^b Accession numbers of full-length cDNA sequences from TIGR (except for *OsBURP07* and *OsBURP11* marked with asterisks)

^c Accession numbers of full-length cDNA sequences from KOME (<http://cdna.01.dna.affrc.go.jp/cDNA/>)

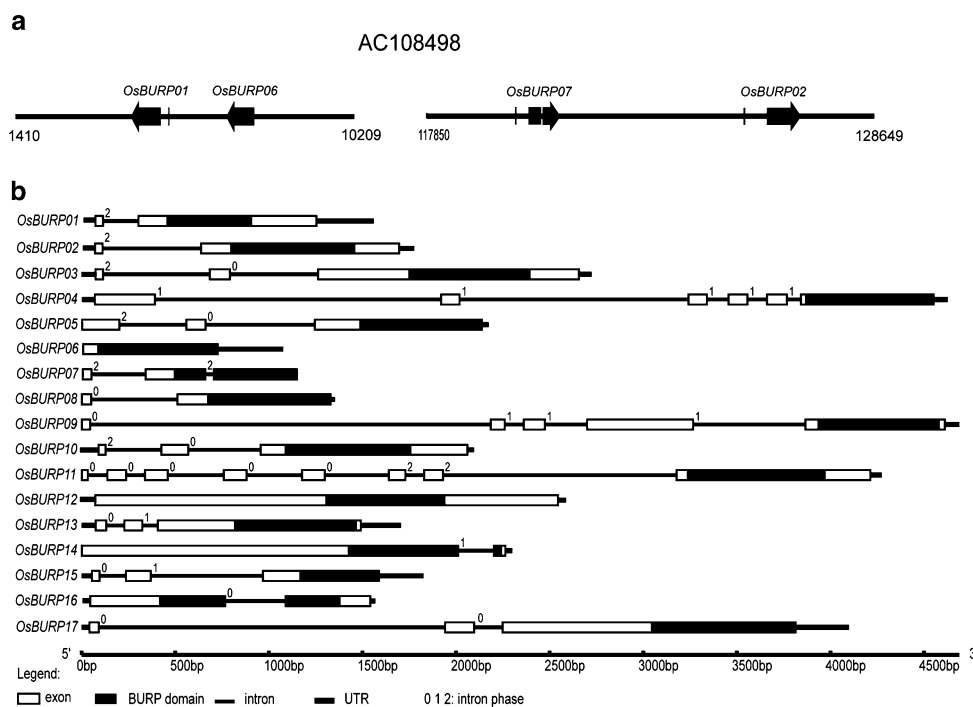
^d Accession number of the BAC/PAC clones and approximate physical distance (Mb) of the chromosome in which the *OsBURP* gene is located

^e The number of exons for putative *OsBURP* genes

^f Chromosome on which the *OsBURP* gene is located

^g PCR efficiencies of primers for real-time PCR

Fig. 1 Genomic organization of *OsBURP* family. **a** Genomic organization of 4 *OsBURP* genes. Tandem arrangement of *OsBURP* in chromosome 5 with its flanking regions. Arrows indicate the direction of each gene. The number indicates the position of the BAC. The white boxes represent the intron and the black boxes represent the exon. **b** Intron–exon organization of the 17 *OsBURP* genes. White rectangles and thin lines represent exons and introns, respectively. Thick lines represent untranslated regions. Black rectangles represent the BURP domains. Numbers denote intron phase



subfamily, the *OsBURP* proteins were distributed in the other five subfamilies. Interestingly, all the members in the BURP V and BURP VI subfamilies are exclusively from rice, i.e., five clustered *OsBURP* genes on chromosome 5 and three clustered *OsBURP* genes on chromosome 6 constitute the two subfamilies. We also noticed that the members of the BURP VII subfamily were all from monocotyledons, while the members of the BNM2-like subfamily and USP-like subfamily are all from dicotyledons. Even in the subfamilies (such as RD22-like and PG1 β -like) containing members from both dicotyledon and monocotyledon species, the members from monocotyledons form a separate branch in the tree and can be easily distinguished from the members from dicotyledons.

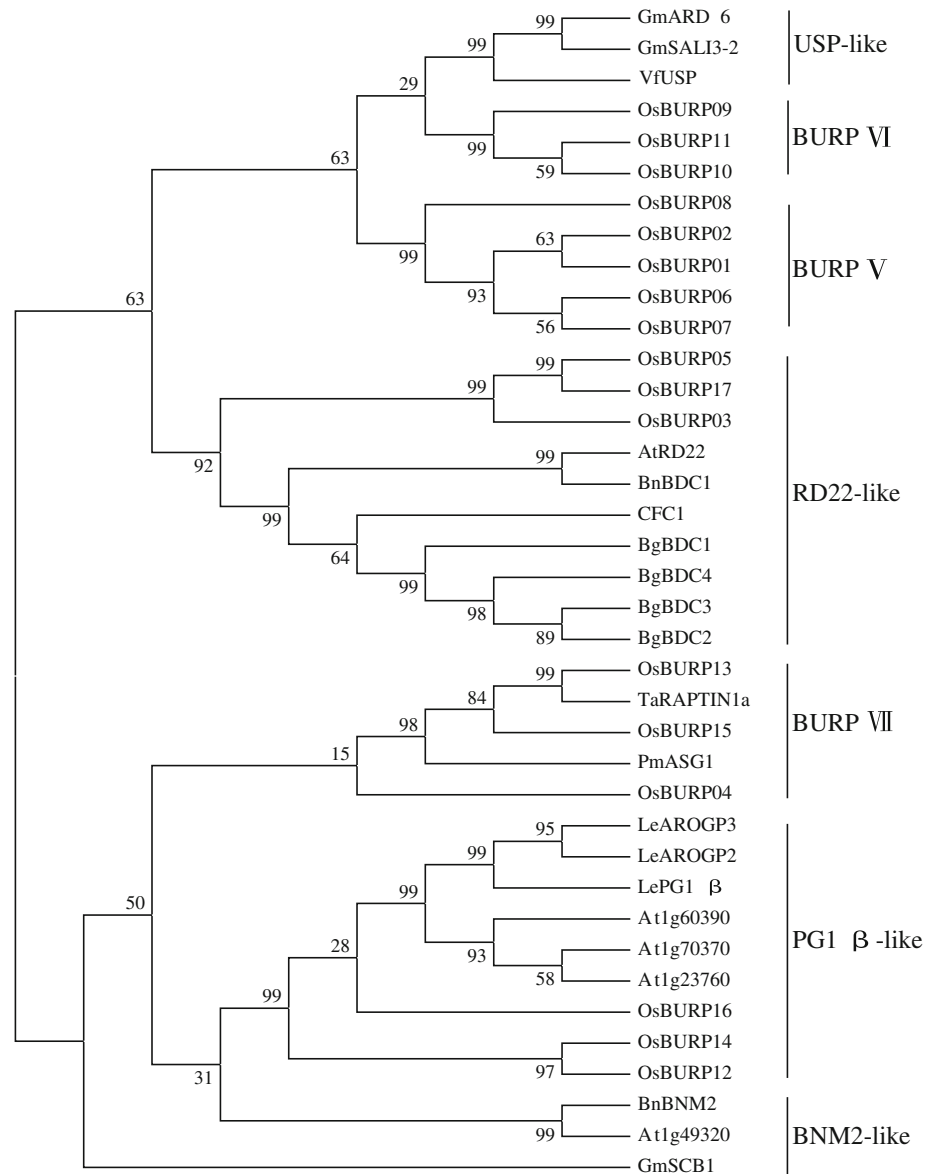
An alignment of the BURP domain sequences revealed several highly conserved amino acid residues: a phenylalanine residue, three glycine residues, two proline residues, a threonine residue, two C residues, and four CH motifs (Fig. 3). But the BURP domains in *OsBURP01* and *OsBURP15* (RA8) were incomplete. *OsBURP01* have two conserved C residues and one CH motif but three repeats of the CH motif are missing. In *OsBURP15*, the two conserved C residues are absent. The distance between two adjacent CH motifs is conserved except for the distance between the second and third, which is longer in the members of the BURP VI subfamily. For example, there are 23 amino acid residues between the second and third CH motif in the members of the BURP V subfamily and 37 amino acid residues in *OsBURP11*. The conserved sequence of the BURP domain can be described as X5–CH–X10–CH–X23–37–CH–X23–26–CH–X8–W.

We further analyzed the conserved motifs in BURP family by MEME version 3.5.7 online (<http://meme.nbcr.net>; Bailey et al. 2006) and 15 conserved motifs were detected (Fig. 4). The sequences of these motifs were used to search against the InterPro database with InterproScan (Mulder and Apweiler 2008). Among 15 conserved motifs, 6 motifs were located within the BURP domain (Fig. 3) for most members of the BURP family. Other motifs have no hits yet in the InterPro database.

Expression levels of *OsBURP* genes in different tissues and organs

Several genes of the *BURP* family have been reported for their differential expression patterns in various tissues and organs. To determine whether the genes of the rice *BURP* family have differential or uniform expression patterns, the transcript levels of all the *OsBURP* genes were investigated in 12 different tissues and organs of rice Minghui 63 by real-time PCR analysis (Fig. 5). The results showed that the *OsBURP* genes exhibit distinct expression patterns in terms of both specificity and expression level. The *OsBURP* genes can be divided into three types according to their expression patterns. The first type of genes (including *OsBURP04*, *OsBURP05*, *OsBURP13*, *OsBURP14*, and *OsBURP15*) had relatively strong expression in the panicle, but their tempo-spatial expression patterns were quite diverse. *OsBURP04* was expressed only in the anther and pistil but not in the young panicle (length less than 2 cm). To the contrary, *OsBURP05* and *OsBURP14* were expressed only in the young panicle but not in the anther

Fig. 2 Phylogenetic relationship of BURP domain-containing proteins from diverse species. The unrooted tree was generated using MEGA3.0 program by the Maximum-likelihood method. The bootstrap values were indicated at the branches. The abbreviations of species names are as follows: At, *Arabidopsis thaliana*; Os, *Oryza sativa*; Vf, *Vicia faba*; Le, *Lycopersicon esculentum*; Bn, *Brassica napus*; Gm, *Glycine max*; Pm, *Panicum maximum*; Bg, *Bruguiera gymnorrhiza*



and pistil. *OsBURP13* and *OsBURP15* were expressed in the young panicle and anther but not in the pistil. The second type of genes (including *OsBURP03*, *OsBURP08*, *OsBURP09*, *OsBURP10*, *OsBURP11*, *OsBURP12*, and *OsBURP16*) showed expression in most of the tissues and organs investigated. *OsBURP10* and *OsBURP11* had relatively higher expression in the root at tillering stage than in other tissues and organs, while *OsBURP09*, which is closely related to *OsBURP10* and *OsBURP11* in the phylogenetic tree, had strong expression in 3-week-old shoots. The third type of genes (including *OsBURP01*, *OsBURP02*, *OsBURP06*, *OsBURP07*, and *OsBURP17*) was expressed in seedlings before tillering stage. *OsBURP02* and *OsBURP17* had strong expression in the shoot at 3 days after germination and in the lamina at the 4-leaf stage, respectively.

Three genes (*OsBURP04*, *OsBURP13*, and *OsBURP15*) of the five panicle-specific genes belong to the BURP VII subfamily. In the gene cluster on chromosome 5, four genes (*OsBURP01*, *OsBURP02*, *OsBURP06*, and *OsBURP07*) located in the same BAC have similar expression patterns in various tissues and organs. They were expressed mainly in seedlings, sheath, and lamina at the 4-leaf stage and have relatively low transcript levels in the stem at tillering stage, but no expression in the panicle. The other gene in this subfamily, *OsBURP08*, was also expressed in seedlings and lamina at the 4-leaf stage but had rather strong expression in the young panicle. All three genes (*OsBURP09*, *OsBURP10*, and *OsBURP11*) of the gene cluster on chromosome 6 were expressed in most of the tissues and organs investigated, suggesting that these genes may have similar or redundant functions.

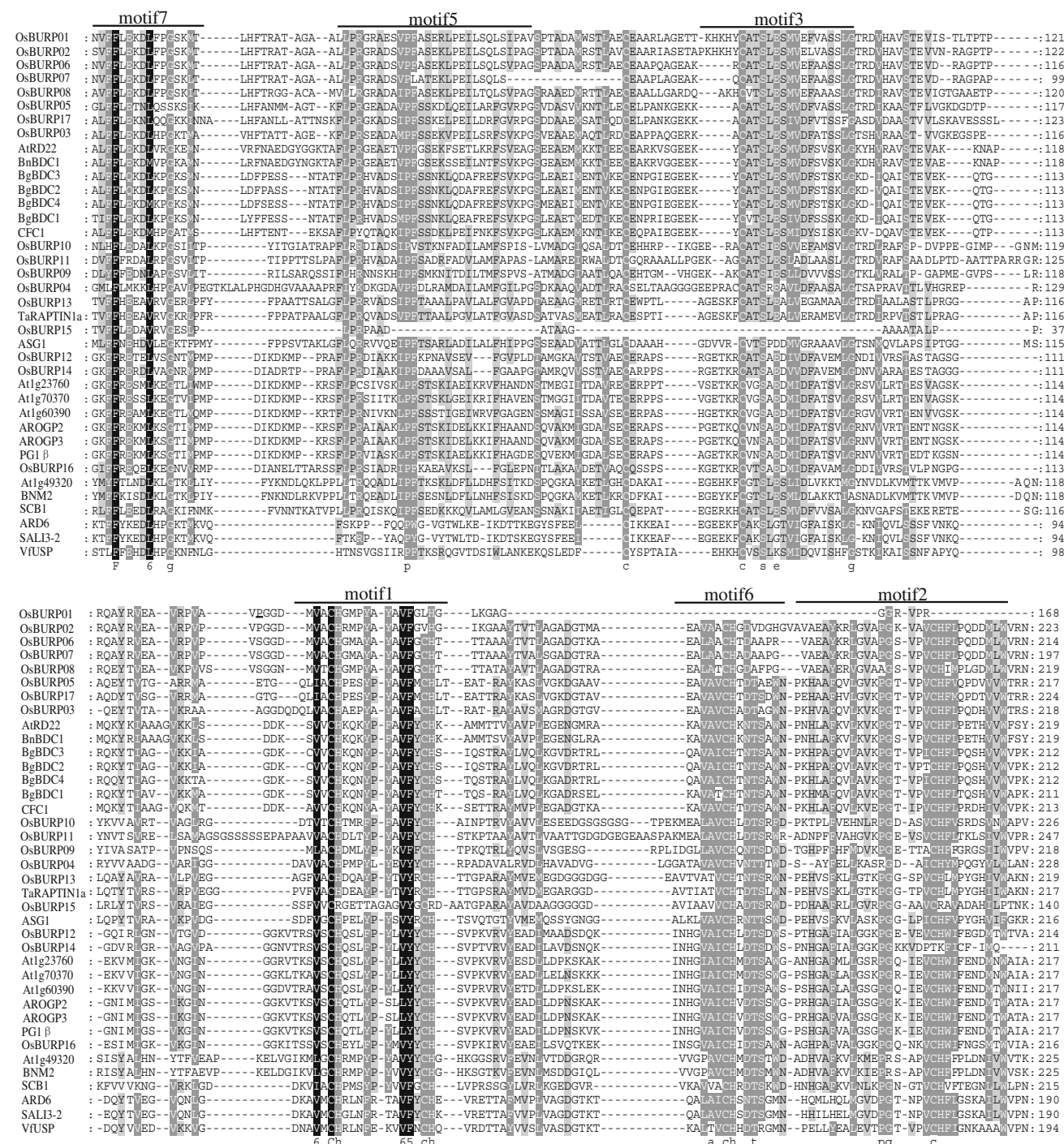


Fig. 3 Alignment of sequences of BURP domain proteins. Motif(n) represent the identified motifs by MEME

Nevertheless, many members from different subfamilies have similar expression patterns. For instance, the *OsBURP05* from the RD22-like subfamily and the *OsBURP14* from the PG1 β -like subfamily and the three genes (*OsBURP04*, *OsBURP13*, and *OsBURP15*) of the BURP VII subfamily were all expressed mainly in the panicle. Seven genes (*OsBURP03*, *OsBURP08*, *OsBURP09*, *OsBURP10*, *OsBURP11*, *OsBURP12*, and *OsBURP16*)

were expressed in most of the tissues or organs, but they were from different subfamilies.

Expression profiles of *OsBURP* genes under stress conditions

A few BURP family genes (such as *RD22*) have been reported for their responsiveness to abiotic stresses. To

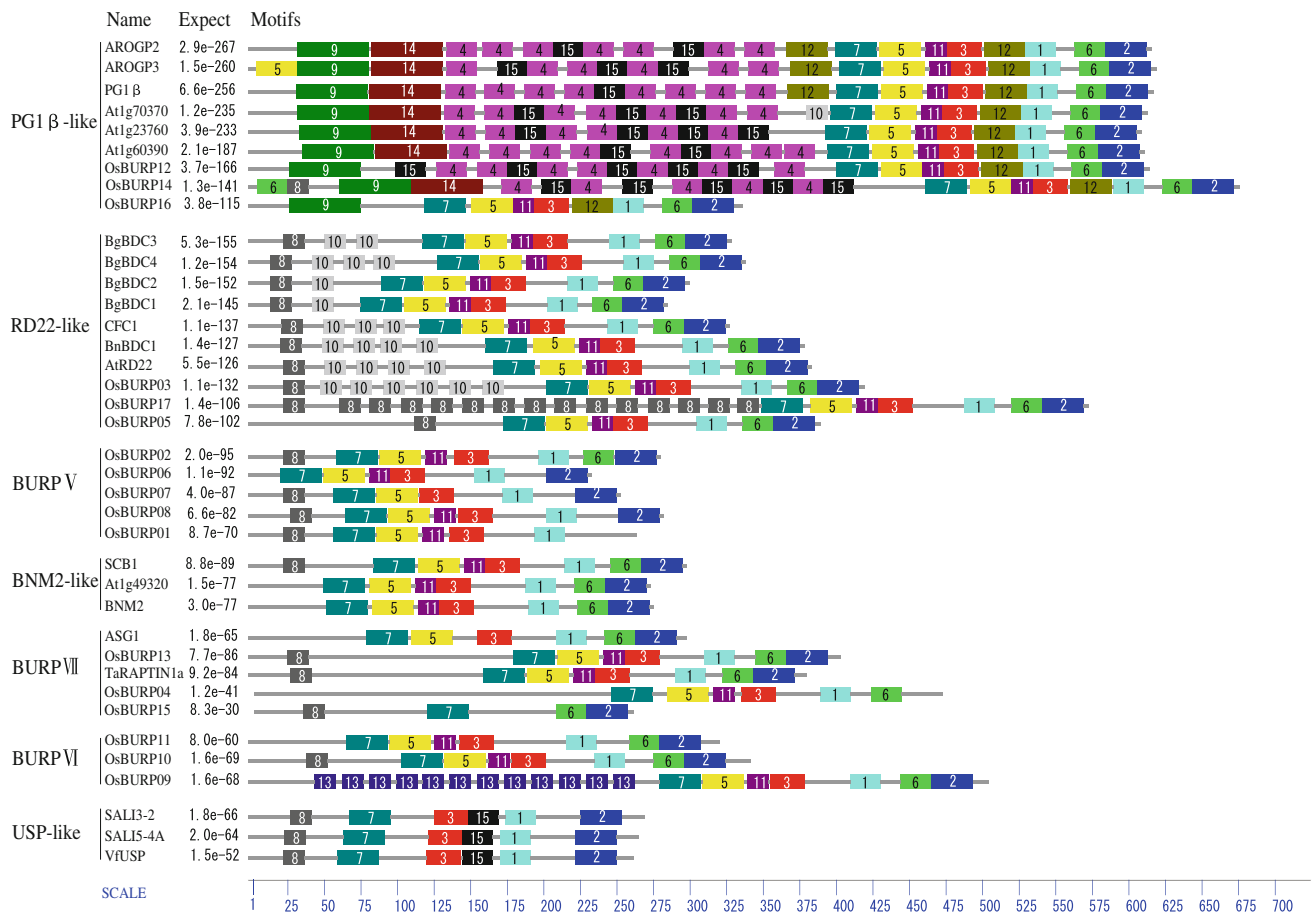


Fig. 4 Fifteen putative motifs identified in the BURP family by MEME software. Different motifs were indicated by different colors. Names of all the members from different subfamilies and combined

P value are shown on the left side of the figure and motif sizes are indicated at the bottom of the figure

obtain a complete view of the stress responsiveness in the rice BURP family, the transcript levels of all the *OsBURP* genes were investigated in seedlings under four treatments including ABA, cold, drought, and salt stresses (Fig. 6). The results showed that 15 *OsBURP* genes were responsive to at least one of the stress treatments applied. Among these genes, five genes (*OsBURP05*, *OsBURP10*, *OsBURP14*, *OsBURP15*, and *OsBURP16*) and five genes (*OsBURP02*, *OsBURP04*, *OsBURP06*, *OsBURP07*, and *OsBURP12*) were up- and down-regulated, respectively, by ABA treatment; four genes (*OsBURP05*, *OsBURP08*, *OsBURP10*, and *OsBURP17*) and one gene (*OsBURP16*) were up- and down-regulated, respectively, by cold treatment; three genes (*OsBURP03*, *OsBURP05*, and *OsBURP16*) and three genes (*OsBURP02*, *OsBURP04*, and *OsBURP12*) were up- and down-regulated, respectively, by drought; and ten genes (*OsBURP04*, *OsBURP05*, *OsBURP08*, *OsBURP09*, *OsBURP10*, *OsBURP11*, *OsBURP14*, *OsBURP15*, *OsBURP16*, and *OsBURP17*) and two genes (*OsBURP06* and *OsBURP12*) were up- and down-regulated, respectively, by salt treatment.

Many *OsBURP* genes were responsive to more than one stress treatment. Among the 10 ABA-responsive genes, 3, 5, and 8 genes were also responsive to cold, drought, and salt stress, respectively. Among the 12 salt-responsive genes, 8, 5, and 4 genes were also responsive to ABA, cold, and drought stress, respectively. Among the 15 stress-responsive *OsBURP* genes, 3 genes (*OsBURP04*, *OsBURP10*, and *OsBURP12*) were responsive to 3 stress treatments and 2 genes (*OsBURP05* and *OsBURP16*) were responsive to 4 stress treatments applied. Interestingly, most genes of the BURP family in rice were responsive to salt treatment. We also noticed that a few genes were responsive mainly to one stress treatment. For example, *OsBURP03* was induced strongly only by drought, but was unresponsive to ABA, cold, and salt treatment.

We compared the real-time PCR results of the BURP family genes with the published or our microarray data for abiotic stresses (Supplemental Table 1), and found some discrepancy between the two sets of data. For example, quite a few genes are not responsive to stresses the in microarray data but actually inducible by stresses based on real-time PCR.

Fig. 5 Tissue-specific expression patterns of *OsBURP* genes. The x-axis is for tissues or organs: *1* root at tillering stage, *2* stems at tillering stage, *3* leaf sheath at 4-leaf stage, *4* leaf lamina at 4-leaf stage, *5* young panicle shorter than 0.2 cm, *6* 0.5–1 cm young panicle, *7* 3–5 cm young panicle, *8* longer than 10 cm young panicle, *9* stamen, *10* pistil, *11* shoot at 3 days after germination, *12* 3-week-old shoot. The y-axis is relative expression level of the gene compared to the expression level in a given tissue. For example, tissue #12, with the relative expression level of the gene *OsBURP01* to *Actin1* normalized as “1” was used as a reference for the relative expression level of in other tissues. Different reference tissues (*underlined* in the x-axis) were used for different genes to illustrate the tissue-specific expression pattern more clearly

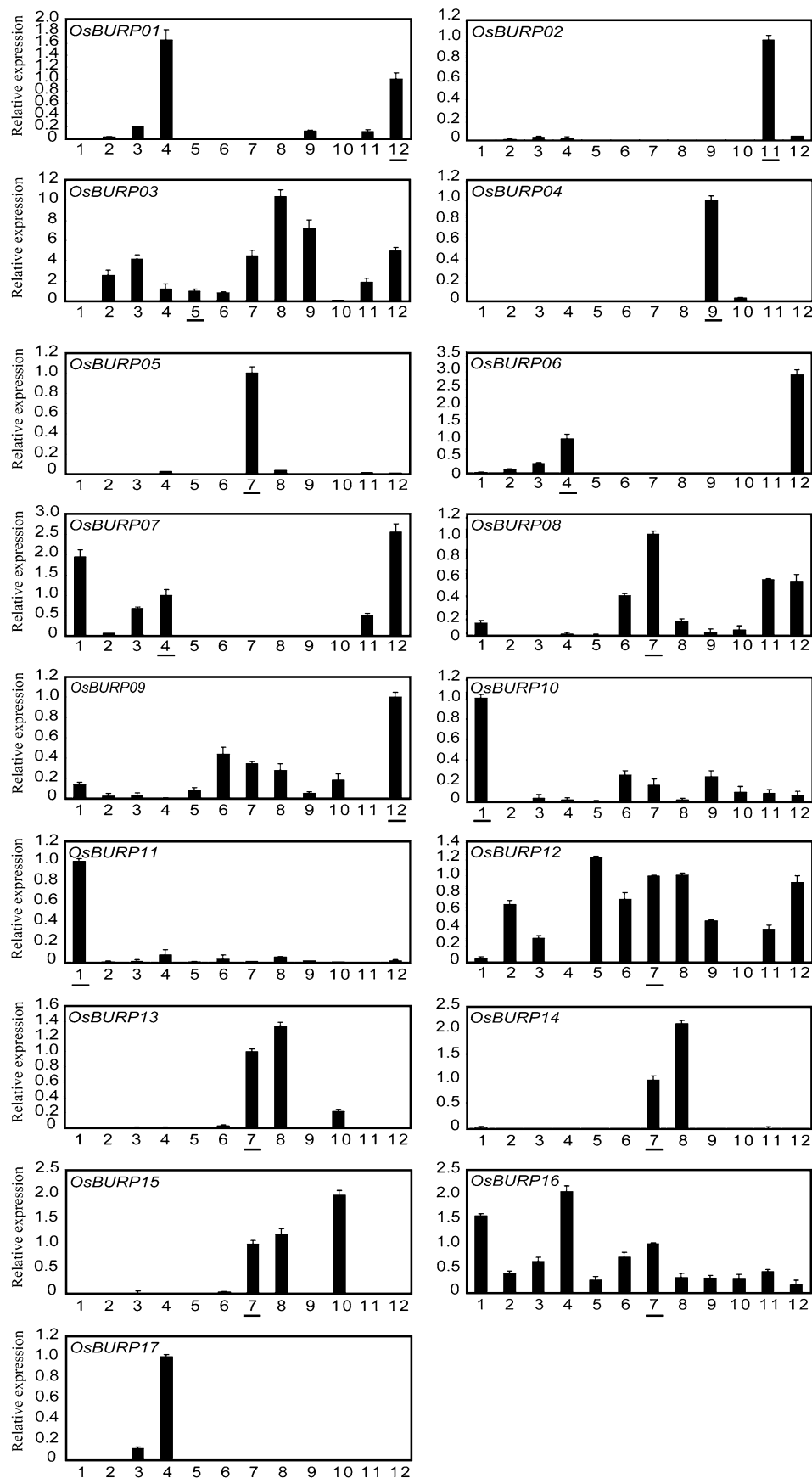


Fig. 6 Expression profiles of *OsBURP* genes under stress conditions. The y-axis is ratio of the relative expression level of the genes to *Actin1* gene under stresses to that under normal conditions. X-axis is time courses of stress treatments. *a* ABA, *c* cold, *d* drought, *s* salt. Stress treatment and time courses were described in the “Materials and methods”. A twofold threshold was used to determine the stress responsiveness of genes

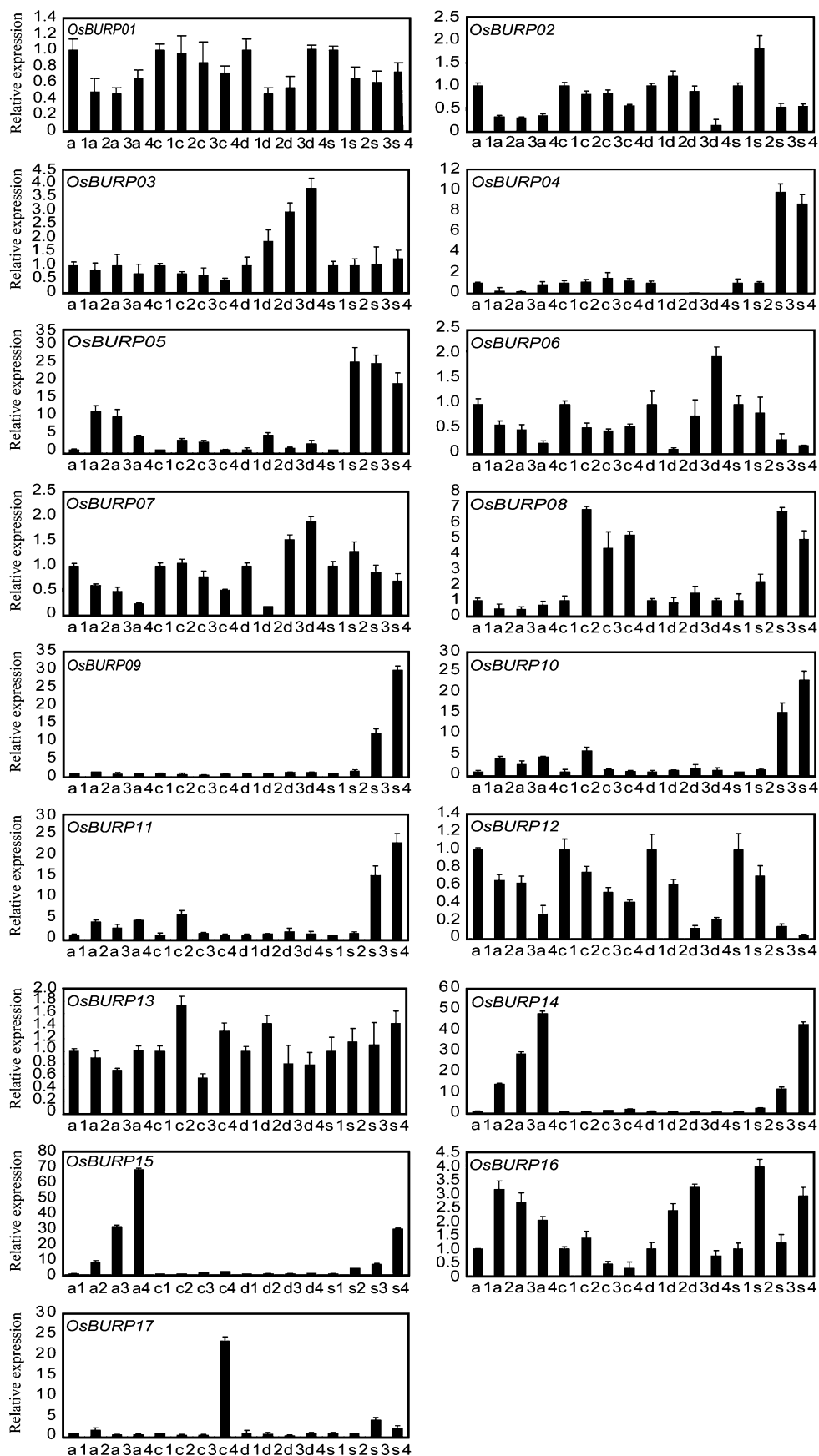
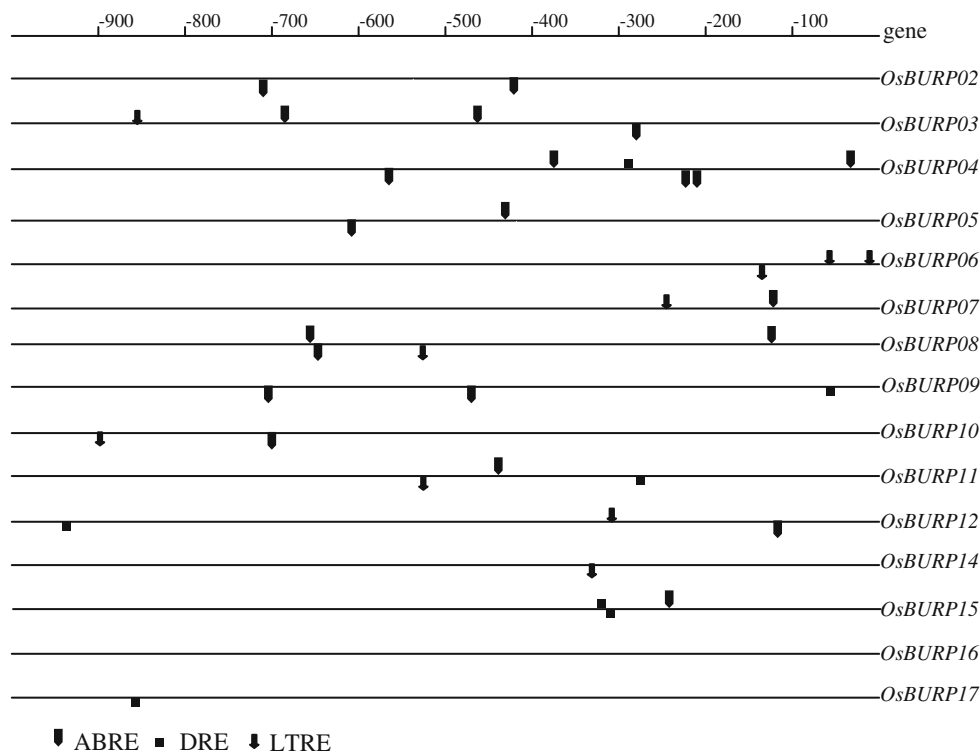


Fig. 7 Analysis of promoter sequences of stress-responsive *OsBURP* genes. Putative ABRE, DRE, and LTRE core sequences in the 1,000 bp promoter regions of the stress-responsive *OsBURP* genes identified by real-time PCR analyses. The lines represent promoter sequences. The elements located in the forward strand (sense strand of the gene) and reverse strand are indicated above and below the lines, respectively



Taken together, these data suggest that the BURP family genes in rice are diversely responsive to abiotic stresses.

In silico analysis of the promoters of *OsBURP* genes

We analyzed the 1,000 bp sequences upstream of the 5' end of the full-length cDNAs for the stress-responsive *OsBURP* genes to identify putative stress-responsive *cis*-elements, such as ABRE (ABA responsive element) (Pla et al. 1993; Narusaka et al. 2003; Yamaguchi-Shinozaki and Shinozaki 2005), DRE (dehydration-responsive element) (Narusaka et al. 2003), and LTRE (low temperature-responsive element) (Yamaguchi-Shinozaki and Shinozaki 1994). Sequence analysis showed that 14 stress-responsive *OsBURP* genes contain putative ABRE, DRE, or LTRE in their promoter regions. Among the ten ABA-responsive *BURP* genes, seven genes contained ABRE in their promoters. However, none of the three types of *cis*-elements was present in the promoter of *OsBURP16*, which was responsive to ABA, cold, drought, and salt treatment (Fig. 7), suggesting that novel stress-responsive *cis*-elements may exist in the promoter of this gene.

Discussion

Structure diversity of the BURP domain-containing proteins

The BURP domain-containing proteins have three or four conserved modules: N-terminal signal peptide, a short seg-

ment, an optional region consisted of repeat unit, and a conserved BURP domain (Hattori et al. 1998). The sequence and number of the repeat unit are different for each member, for example, FXXY in PG1 β , TXV in RD22, and a few copies of TXVGVGXXGVXVDAGKXX in 4 BgBDCs (Zheng et al. 1992; Yamaguchi-Shinozaki and Shinozaki 1993; Banzai et al. 2002). The region between the signal peptide and the BURP domain is the most obvious structure distinguishing the members of the BURP family, and the BURP domain is strikingly similar among members of the BURP family. Within this domain, many amino acid positions are highly conserved, including 2 C residues and four repeats of the CH motif. The distance between two adjacent CH motifs is conserved except that between the second and third. For example, in the three members of the BURP VI subfamily, the distances between the second CH motif and third are longer than that of other BURP domain-containing proteins. There are 37 amino acids between the second and third CH motifs in *OsBURP11*, but this distance is only 25–26 amino acids for other BURP domain-containing proteins except for the members of the BURP VI subfamily. Through alignment, the conserved sequence of the BURP domain can be described as X5-CH-X10-CH-X23-37-CH-X23-26-CH-X8-W.

Our understanding is very limited with regard to the function of each module. The hydrophobic N-terminal region may act as a signal peptide for transport into the endoplasmic reticulum. The signal peptide is cleaved from PG1 β in the translation process (Zheng et al. 1992) and affects the function of the BURP protein; for instance,

deletion of the signal peptide in SAIL-2 protein can reduce the tolerance of transgenic yeast to salt (Tang et al. 2007). Although the *BURP* genes share similar sequence and primary structures, their expression patterns in various tissues and organs and under various stress conditions are different. So the most obvious region, including a short and nonrepetitive segment and an optional repeat segment, may be crucial for their distinct functions. But the short and nonrepetitive segment may be considered simply as a kind of stuffer fragment in BNM2 and VtUSP (Bassünér et al. 1988). The region following the N-terminal signal peptide may have a delivery function to transport the proteins to a cellular target compartment; for instance, the cell wall for PG1 β . Regarding the function of the BURP domain, a previous report suggested that the BURP domain-containing proteins function as polypeptides or may be involved in targeting attached polypeptides to a defined subcellular location (Hattori et al. 1998). The BURP domain-containing proteins may exert a common function because of the highly similar BURP domain. The various and attached regions may endue the BURP domain-containing proteins with unique functions. The BURP domain of SALI3-2 is crucial for its tolerance to salt (Tang et al. 2007). The PG1 β can form a PG1 complex with PG2 and is involved in the immobilization of PG2 at the cell wall and regulation of its activity (Zheng et al. 1992). Through the BURP domain, SCB1 is localized on the cell wall and interacts with other interrelated proteins (Batchelor et al. 2002). Based on the primary amino acid sequence of the BURP domain, the BURP domain may integrate with metal ions, for example, Cu²⁺, Ag⁺, and so on, with CH motifs.

Evolution of *BURP* family genes

The *BURP* genes can be divided into seven subfamilies based on the phylogenetic analysis. Among the seven subfamilies, three subfamilies (BURP V, BURP VI, and BURP VII) were composed of *BURP* genes all from monocotyledons and two subfamilies (USP-like and BNM2-like) were composed of *BURP* genes all from dicotyledons. In the other two mixed subfamilies (RD22-like and PG1 β -like), the *BURP* genes from monocotyledons and dicotyledons were distinctly different. In the RD22-like subfamily, members of rice are differentiated from those of *Arabidopsis*, cotton, and mangrove. This suggested that the functions and evolution of the BURP genes may have some variation in monocotyledons and dicotyledons. The *BURP* genes might have originated before the divergence of monocotyledons and dicotyledons and then evolved separately.

Based on the primary structures of the BURP proteins, Hattori et al. (1998) suggested that the shuffling of the signal peptide and C-terminal BURP domain and the amplifi-

cation of a short basic unit of the optional repeated sequence play a role in the evolution of BURP genes. The signal peptide is present in almost all putative BURP domain-containing proteins, and the sequences of the signal peptide and BURP domain are very similar to each other. However, the regions following the signal peptide vary by the family (Hattori et al. 1998). This feature provides evidence for the exon shuffling model. According to this model, many genes were created through “mixing and matching” of basic functional exon units and the formation of a *BURP* gene needs at least four exons to be joined together. But few *BURP* genes meet this requirement in rice. So the amplification of a short basic unit and duplication of gene may be more essential for the evolution of BURP genes than the exon shuffling. This can be supported by the following: (1) the conserved signal peptide in almost all members of BURP domain-containing proteins; (2) an optional segment consisting of repeated units that are unique to each member and the differential number of the same repeated units in each subfamily, such as the RD22-like subfamily; (3) the sequence similar genes, including the repeat unit, on the same chromosome, for example, the members of the BURP V and BNM2-like subfamily in rice; (4) 16 out of 18 repeat units being closely related to each other in PG1 β and the other 2 repeat units being incomplete, possibly formed by unequal crossing; (5) the incomplete BURP domain in OsBURP01 and OsBURP15 possibly being formed by unequal cross-over; (6) 4 *OsBURP* genes in the same BAC.

Diverse expression profiles of *OsBURP* genes

In this study, we analyzed the expression patterns of *OsBURP* genes in different tissues and organs and under various stress conditions and noticed that many genes with similar sequences have similar expression patterns. For instance, both *OsBURP06* and *OsBURP07*, which have high sequence similarity, are expressed in seedlings before tillering stage and are responsive to ABA. The transcript levels of four genes (*OsBURP01*, *OsBURP02*, *OsBURP06*, and *OsBURP07*, the five clustered genes on chromosome 5) of the BURP V subfamily showed similar expression patterns in different tissues and organs and under different stress conditions. They were all expressed mainly in the seedlings before tillering, but not in the panicle. The similar expression patterns of closely related genes may suggest that they have redundant functions. On the contrary, we also noticed that many genes with similar sequences have different expression patterns in different tissues and organs and under various stress conditions. For example, *OsBURP05* and *OsBURP17* were grouped very closely in the phylogenetic tree, but their expression patterns were sharply different. *OsBURP05* was expressed mainly in the

panicle and was induced by four stress treatments, including ABA, cold, drought, and salt, but *OsBURP17* was expressed in the leaf sheath and lamina at the four-leaf stage and was only induced by cold and salt stresses. Their biochemical functions may diverge along with the evolution of each gene after a duplication event. We also noticed the stress induced expression patterns of some *BURP* genes from different subfamilies are similar. For example, both *OsBURP08* and *OsBURP17*, from the BURP V and RD22-like subfamilies, respectively, were induced by cold and salt, but were not responsive to ABA and drought. *OsBURP05*, *OsBURP10*, and *OsBURP16*, which are from different subfamilies, were induced by at least three stress treatments.

The members of the BURP VII subfamily, containing *OsBURP04*, *OsBURP13*, and *OsBURP15*, are all from monocotyledons and all have a high transcript level in flower organs. But their tempo-spatial expression patterns are different. For instance, *TaRAPTIN1a* and *ASG1* were anther-specific and apomixis-specific, respectively (Wang et al. 2003; Chen et al. 2005). *OsBURP13* (*OsRAPTIN1*) and *OsBURP15* (*RA8*) had been reported as anther-specific genes (the same as our result) and had similar expression profiling in the young panicle shorter than 0.2 cm and anther but not in the pistil (Jeon et al. 1999; Wang et al. 2003). *OsBURP04* was expressed mainly in the pistil and very weakly in the anther, but was not expressed in the young panicle. But our understanding of their function was very limited except for *OsBURP13*. *OsBURP13* may play an important role in male fertility by regulating transport of sporopollenin from the tapetum to developing microspores via Ubisch bodies (Wang et al. 2003).

Involvement of *BURP* family genes in stress responses

To date, the functions of most of the *BURP* family members are unknown. The induced expression of many *OsBURP* genes in different stress conditions provided strong evidence of the involvement of this family in tolerance to stress. The members of the RD22-like subfamily, containing *OsBURP03*, *OsBURP05*, and *OsBURP17*, were induced by at least one stress treatment, and both ABA-dependent and ABA-independent pathways were important for stress tolerance. For example, *RD22*, a drought-responsive gene, was mediated by ABA in *Arabidopsis* (Yamaguchi-Shinozaki and Shinozaki 1993). The transcript level of *BnBDC1*, a shoot-specific gene from oilseed rape, was up-regulated by salt, ABA, and osmotic stress and down-regulated by salicylic acid, but it was unresponsive to cold treatment (Yu et al. 2004). Four genes encoding BURP domain-containing proteins (*BgBDC1*, 2, 3, and 4) in the mangrove and showing similar sequences to *RD22* were all induced by salt, ABA, and drought stress (Banzai et al.

2002). *CFC1* was defined as a dehydration-induced protein RD22-like protein from cotton, but its function was unknown. *OsBURP03* was induced only by drought and was unresponsive to ABA, cold, and salt. *OsBURP05* was induced by all four stress conditions mentioned above. The transcript level of *OsBURP17* was strongly up-regulated by cold and weakly by salt. Although all members of the RD22-like subfamily were responsive to at least one stress treatment, the mechanisms in response to stress might be different. *RD22*, *BnBDC1*, *BgBDCn* ($n = 1, 2, 3$, and 4), and *OsBURP05* were responsive to ABA and at least other one stress treatment, so their responses to stresses may be mediated by ABA. *OsBURP03* was induced by drought and *OsBURP17* was induced by cold and salt, and neither was responsive to ABA and they may respond to the stresses through an ABA-independent pathway.

Many members of BURP family in rice are tissue-specific and also induced by stresses. For example, *RD22* was expressed in flowers and the expression of *RD22* was induced by salt and ABA stress as well as water deficit but not cold stress. *OsBURP05* was strongly expressed in pistil and up-regulated by ABA, cold, drought, and salt stresses. *OsBURP11* was strongly expressed in root and induced by salt. These results suggest that some of these *OsBURP* genes may be important both for developmental processes and in response to abiotic stresses.

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